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Power state management for fabrics

Abstract

In one embodiment, a retimer device includes a receiver to receive data from a first device via a data link, retimer circuitry to recover a clock phase from the received data, and prepare a new copy of the received data sampled by a clean clock based on the recovered clock phase, a transmitter to transmit the new copy to a second device via the data link, wherein the receiver is configured to receive an in-band standby signal from the first device having a given pattern in a physical layer of the signal, activate a power saving mode of the retimer device responsively to the standby signal having the given pattern in the physical layer of the standby signal, receive an in-band wakeup signal from the first device, and initiate an exit from the power saving mode to power up the retimer device responsively to the wakeup signal.

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Background/Summary

RELATED APPLICATION INFORMATION (1) The present application claims benefit of U.S. Provisional Patent Application Ser. 63/392,879 of Meltser, et al., filed 28 Jul. 2022, the disclosure of which is hereby incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention relates to computers, and in particular, but not exclusively to, power state management in links.

BACKGROUND

(2) Power state management is used to save power in networks and connections between devices. For example, two devices either side of a passive link may perform a handshake to go into a power saving mode (e.g., L1 power saving mode) to save energy when the link is not actively being used to transfer packets or other data. Power state management may also be used to throttle performance to other chip computation engines by maintaining a constrained thermal envelope.

SUMMARY

(3) There is provided in accordance with an embodiment of the present disclosure, a retimer device, including a receiver to receive data from a first device via a data link, retimer circuitry to recover a clock phase from the received data, and prepare for transmission a new copy of the received data sampled by a clean clock based on the recovered clock phase, a transmitter to transmit the new copy of the received data to a second device via the data link, wherein the receiver is configured to receive an in-band standby signal from the first device having a given pattern in a physical layer of the signal, activate a power saving mode of the retimer device so that at least some of the retimer device is powered down responsively to the standby signal having the given pattern in the physical layer of the standby signal, receive an in-band wakeup signal from the first device, and initiate an exit from the power saving mode to power up the retimer device responsively to the wakeup signal.

(4) Further in accordance with an embodiment of the present disclosure the receiver is configured to activate the power saving mode of the retimer circuitry so that the retimer circuitry is powered down responsively to the in-band standby signal, and initiate an exit from the power saving mode to power up the retimer circuitry responsively to the wakeup signal.

(5) Still further in accordance with an embodiment of the present disclosure the receiver recognizes the given pattern in the standby signal in a physical media attachment sub layer without needing to use a physical coding sub layer.

(6) Additionally in accordance with an embodiment of the present disclosure the transmitter is configured to send another given pattern in the physical layer of another in-band standby signal to the second device to activate a power saving mode in the second device.

(7) Moreover, in accordance with an embodiment of the present disclosure the transmitter is configured to send the wakeup signal in-band to the second device responsively to receiving the wakeup signal from the first device.

(8) Further in accordance with an embodiment of the present disclosure the transmitter is configured to send the wakeup signal to the second device prior to, or while, the retimer device is powering up from the power saving mode.

(9) Still further in accordance with an embodiment of the present disclosure the transmitter is configured to generate another wakeup signal and send the other wakeup signal in-band to the second device responsively to receiving the wakeup signal from the first device.

(10) Additionally in accordance with an embodiment of the present disclosure the data link includes an electrical signal data link.

(11) Moreover, in accordance with an embodiment of the present disclosure the data link includes an optical signal data link.

- (12) Further in accordance with an embodiment of the present disclosure the first device is an end device, and the second device is an end device.
- (13) Still further in accordance with an embodiment of the present disclosure the first device is a retimer device and the second device is an end device.
- (14) Additionally in accordance with an embodiment of the present disclosure the first device is an end device, and the second device is a retimer device.
- (15) Moreover, in accordance with an embodiment of the present disclosure the first device is a retimer device and the second device is a retimer device.
- (16) There is also provided in accordance with another embodiment of the present disclosure, a retimer device power saving mode method, including receiving data from a first device via a data link, recovering a clock phase from the received data, preparing for transmission a new copy of the received data sampled by a clean clock based on the recovered clock phase, transmitting the new copy of the received data to a second device via the data link, receiving an in-band standby signal from the first device having a given pattern in a physical layer of the signal, activating a power saving mode of a retimer device so that at least some of the retimer device is powered down responsively to the standby signal having the given pattern in the physical layer of the standby signal, receiving an in-band wakeup signal from the first device, and initiating an exit from the power saving mode to power up the retimer device responsively to the wakeup signal.
- (17) Further in accordance with an embodiment of the present disclosure the activating includes activating the power saving mode of retimer circuitry so that the retimer circuitry is powered down responsively to the in-band standby signal, and initiating an exit from the power saving mode to power up the retimer circuitry responsively to the wakeup signal.
- (18) Still further in accordance with an embodiment of the present disclosure, the method includes recognizing the given pattern in the standby signal in a physical media attachment sub layer without needing to use a physical coding sub layer.
- (19) Additionally in accordance with an embodiment of the present disclosure, the method includes sending another given pattern in the physical layer of another in-band standby signal to the second device to activate a power saving mode in the second device.
- (20) Moreover, in accordance with an embodiment of the present disclosure, the method includes sending the wakeup signal in-band to the second device responsively to receiving the wakeup signal from the first device.
- (21) Further in accordance with an embodiment of the present disclosure the sending includes sending the wakeup signal to the second device prior to, or while, the retimer device is powering up from the power saving mode.
- (22) Still further in accordance with an embodiment of the present disclosure, the method includes generating another wakeup signal and sending the other wakeup signal in-band to the second device responsively to receiving the wakeup signal from the first device.
- (23) Additionally in accordance with an embodiment of the present disclosure the data link includes an electrical signal data link.
- (24) Moreover, in accordance with an embodiment of the present disclosure the data link includes an optical signal data link.
- (25) Further in accordance with an embodiment of the present disclosure the first device is an end device, and the second device is an end device.
- (26) Still further in accordance with an embodiment of the present disclosure the first device is a retimer device and the second device is an end device.
- (27) Additionally in accordance with an embodiment of the present disclosure the first device is an end device, and the second device is a retimer device.
- (28) Moreover, in accordance with an embodiment of the present disclosure the first device is a retimer device and the second device is a retimer device.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present invention will be understood from the following detailed description, taken in conjunction with the attached Annex and the drawings in which:
- (2) FIG. 1 is a block diagram view of a power state management system constructed and operative in accordance with an embodiment of the present invention;
- (3) FIG. 2 is a flowchart including steps in a power-down method in the system of FIG. 1;
- (4) FIG. 3 is flowchart including steps in a power-up method in the system of FIG. 1;
- (5) FIG. 4 is a block diagram view of a retimer device for use in the system of FIG. 1; and
- (6) FIG. 5 is a block diagram of a retimer integrated circuit of the retimer device of FIG. 4.

DESCRIPTION OF EXAMPLE EMBODIMENTS

Overview

- (7) As previously mentioned, power state management is used to save power in links between devices. For example, two devices either side of a passive link may perform a handshake to take both devices into a power saving mode to save energy when the link is not actively being used to transfer packets or other data.
- (8) When dealing with active links including such components as retimer devices, power saving management is more complex. A typical network/Ethernet retimer device is a mixed-signal analog/digital device that is not protocol aware and has the ability to recover a clock phase from a data stream, and prepare and retransmit a fresh copy of the data sampled by a clean clock, which is based on the recovered clock phase. The term “clean clock” as used in the specification and claims is defined to include a “recovered clock” recovered by a clock data recovery (CDR) function, which may compensate for phase-delay variations and jitter.
- (9) The retimer device may include a continuous time linear equalizer (CTLE), a variable gain amplifier (VGA), driver stages, a clock data recovery (CDR) circuit, a long-tail equalizer (LTE), and a decision feedback equalizer (DFE). The LTE compensates for long-term impulse response impairments, and the DFE acts as a nonlinear equalizer, suppressing the ISI due to channel imperfections such as high-frequency losses and notches. Internal digital logic, state machines, and/or a microcontroller may manage the automatic adaptation of the CTLE, VGA, LTE, and DFE blocks, and implement protocol link training and status updates.
- (10) One solution to allowing retimer devices to be placed in power saving mode is to have full physical coding sub-layer (PCS) state machines in each retimer device so that the retimer devices in the link can be placed in power saving mode via a handshake from one or more other devices at either end of the link. The above solution is impractical and adds too much complexity to the retimer devices as well as taking too long to power down and power up the retimer devices. Other solutions may include using out-of-band signaling to signal the retimer devices to enter and exit power saving mode.
- (11) Embodiments of the present invention solve at least some of the above problems by allowing a retimer device (or devices) in an active link between two end devices (e.g., host devices, chips or nodes, or any combination thereof) at either end of the link to be signaled using an in-band standby signal (sent from an end device or another retimer device) to place the retimer into a power saving mode and using an in-band wakeup signal (sent from the end device or another retimer device) to cause the retimer to exit from the power saving mode.
- (12) The in-band standby signal may include a pseudo-random pattern in the physical signal which is recognized by the retimer device. The given pattern may be recognized and processed by a physical media attachment (PMA) sub-layer without having to use PCS state machines. The receiving retimer device generally also synchronizes to, and locks on to the pseudo-random pattern in a noisy environment and/or when receiving the pattern from a noisy channel. The length of the

pattern is generally sufficiently long to ensure that the pattern is unlikely to be falsely identified as part of resiliency to any potential bit flips or receiver impairments. Protecting against false identification may be important as the PMA sub-layer is generally not protected by a forward error correction (FEC) mechanism. The retimer device enters power saving mode upon receipt of the standby signal including the given pattern. Entering power saving mode may include powering down retimer circuitry such as circuitry used for retiming data, modulation, and equalization. The retimer device may then also transmit the given pattern in an in-band physical signal to the next electrical or optical device (e.g., retimer device or end device) in the link so that the next element may enter power saving mode, etc. in a cascaded fashion until all retimer devices along the link are in power saving mode.

(13) Upon the retimer device(s) in the link entering power saving mode the link is maintained in zero differential, i.e., common mode on both p/n wires. For example, both wires of the link (p and n, if an electrical wire is used) are maintained at the same voltage. For an optical link, the optical link is maintained in common mode without modulation, for example.

(14) When one of the end devices at either end of the link determines that the link should be brought out of power saving mode, the end device transmits an in-band wakeup signal to the first retimer device in the link. The wakeup signal may take the form of a specific signal or any signal which takes the link out of common mode or zero differential, for example, by providing a wakeup signal in a given frequency range and with a minimum differential swing. The wakeup signal could be a clock pattern or random pattern, for example. The wakeup signal is detected by a receiver of the first retimer device (e.g., based on the link exiting common mode) and causes the first retimer device to exit power saving mode and transmit a wakeup signal (e.g., the same received wakeup signal or a new wakeup signal) to the next device (e.g., retimer device or end device) in the link etc. in a cascaded fashion until all retimer devices along the link have exited power saving mode. The wakeup signal may be sent to the next device, prior to, or while, the first retimer device is exiting power save mode.

(15) The above may be used to cause retimer devices to enter and exit power saving mode for electrically based links (e.g., copper wire or PCB trace links) or optical based links (e.g., using optical fibers).

System Description

(16) Reference is now made to FIG. 1, which is a block diagram view of a power state management system **10** constructed and operative in accordance with an embodiment of the present invention. The power state management system **10** includes a plurality of retimer devices **12** on an active data link **14** between two end devices **16**. One of the retimer devices **12**, retimer device **12-1**, is shown in an inset **18** of FIG. 1. Part of one of the end devices **16**, an end device **16-1**, and part of one of another one of the retimer devices **12**, retimer device **12-2**, are also shown in inset **18**. The active data link **14** also includes a third retimer device **12-3**. Three retimer devices **12** are shown by way of example only, the active data link **14** may include any suitable number of retimer devices **12**, such as one, two, three or more retimer devices **12**.

(17) The retimer devices **12** are described in more detail by way of retimer device **12-1** shown in more detail in inset **18**. The retimer devices **12-1** includes a receiver **20**, a transmitter **22**, and retimer circuitry **24**.

(18) In normal operation, the receiver **20** is configured to receive data (e.g., packets) from the end device **16-1** via the active data link **14**. The retimer circuitry **24** is configured to recover a clock phase from the received data and prepare for transmission a new copy of the received data sampled by a clean version of the recovered clock phase. The transmitter **22** is configured to transmit the new copy of the received data to the retimer device **12-2** via the data link **14**. If the active link **14** does not include the retimer devices **12-2**, **12-3**, then the transmitter **22** is configured to transmit the new copy of the received data to the end device **16-2**. The active link **14** may include any suitable data link, for example, an electrical signal data link and/or an optical signal data link. Depending

on the number of retimer devices **12** in the active data link **14** and the position of the retimer device **12-1** in the active data link **14**, in other embodiments, the retimer device **12-1** may receive data from one of the end devices **16** or one of the retimer devices **12**, and transmit data to one of the end devices **16** or one of the retimer devices **12**.

(19) Reference is now made to FIG. **2**, which is a flowchart **200** including steps in a power-down method in the system **10** of FIG. **1**. Reference is also made to FIG. **1**. Entering power saving mode is now described in more detail.

(20) The receiver **20** is configured to receive an in-band standby signal from the end device **16-1** (block **202**). The standby signal has a given pattern in a physical layer of the standby signal. The receiver **20** is configured to activate a power saving mode (e.g., L1 power saving mode) of the retimer device **12-1** so that at least some of the retimer device **12-1** is powered down responsively to the standby signal having the given pattern in the physical layer of the standby signal (block **204**). In some embodiments, the receiver **20** recognizes the given pattern in the standby signal in a physical media attachment (PMA) sub layer without needing to use a physical coding sublayer (PCS) state machine. In some embodiments, the receiver **20** is configured to activate a power saving mode of the retimer circuitry **24** so that the retimer circuitry **24** is powered down responsively to the in-band standby signal.

(21) In some embodiments, the transmitter **22** is configured to send the same given pattern (as received by the receiver **20**) in the physical layer of the same standby signal to the retimer device **12-2** (or to the end device **16-2** if the active data link **14** does not include the retimer devices **12-2**, **12-3**) to activate a power saving mode in the retimer device **12-2** (or to the end device **16-2** if the active data link **14** does not include the retimer devices **12-2**, **12-3**) (block **206**).

(22) In other embodiments, the transmitter **22** is configured to generate another standby signal and send another given pattern in the physical layer of the other in-band standby signal to the retimer device **12-2** (or to the end device **16-2** if the active data link **14** does not include the retimer devices **12-2**, **12-3**) to activate a power saving mode in the retimer device **12-2** (or to the end device **16-2** if the active data link **14** does not include the retimer devices **12-2**, **12-3**) (block **208**).

(23) The standby signal (or equivalent thereof) is propagated by each device to the next device in the active data link **14** in a cascaded fashion thereby causing each device in the active data link **14** to enter the power saving mode.

(24) Reference is now made to FIG. **3**, which is flowchart **300** including steps in a power-up method in the system **10** of FIG. **1**. Reference is also made to FIG. **1**. Exiting power saving mode is now described in more detail.

(25) The receiver **20** is configured to receive an in-band wakeup signal from the end device **16-1** (block **302**) and initiate an exit from the power saving mode to power up the retimer device **12-1** responsively to the wakeup signal (block **304**). In some embodiments, the receiver **20** is configured to initiate an exit from the power saving mode to power up the retimer circuitry **24** responsively to the wakeup signal. The transmitter **22** is configured to send the wakeup signal in-band to the retimer device **12-2** (or to the end device **16-2** if the active data link **14** does not include the retimer devices **12-2**, **12-3**) responsively to receiving the wakeup signal from the end device **16-1** (block **306**). The transmitter **22** is configured to send the wakeup signal to the retimer device **12-2** (or to the end device **16-2** if the active data link **14** does not include the retimer devices **12-2**, **12-3**) prior to, or while, the retimer device **12-1** is powering up from the power saving mode. In other embodiments, the transmitter **22** is configured to generate another wakeup signal (block **308**) and send the other wakeup signal in-band to the retimer device **12-2** (or to the end device **16-2** if the active data link **14** does not include the retimer devices **12-2**, **12-3**) responsively to receiving the wakeup signal from the retimer device **12-1** (block **310**).

(26) The wakeup signal (or equivalent thereof) is propagated by each device to the next device in the active data link **14** in a cascaded fashion thereby causing each device in the active data link **14** to exit the power saving mode.

(27) In practice, some or all of the functions of the receiver **20** and/or the transmitter **22** may be combined in a single physical component or, alternatively, implemented using multiple physical components. These physical components may comprise hard-wired or programmable devices, or a combination of the two. In some embodiments, at least some of the functions of the receiver **20** and/or the transmitter **22** may be carried out by a programmable processor under the control of suitable software. This software may be downloaded to a device in electronic form, over a network, for example. Alternatively, or additionally, the software may be stored in tangible, non-transitory computer-readable storage media, such as optical, magnetic, or electronic memory.

(28) Reference is now made to FIGS. **4** and **5**. FIG. **4** is a block diagram view of an example retimer device, which may be one of the retimer devices **12** for use in the system **10** of FIG. **1**. Other retimer devices may be used in the system **10**. FIG. **5** is a block diagram of an example retimer integrated circuit **26** of the retimer device **12** of FIG. **4**. FIGS. **4** and **5** shows various elements of the retimer device **12**. The retimer devices **12** includes the retimer integrated circuit **26**, optical transimpedance amplifiers **28**, an externally modulated laser **30**, one or more photodiodes **32**, flash or EEPROM memory **36**, a micro-controller **38**, supply regulators **40**, and interfaces **42**. In non-optical retimer devices optical-specific elements are typically missing.

(29) Most of the power saving in the retimer device **12** is in the retimer integrated circuit **26** shown in more detail in FIG. **5**. The retimer integrated circuit **26** includes a crossbar **44**, a Serializer/Deserializer (SerDes) and phase-locked loop (PLL) **46**, a boot clock, DFX (a term used to describe an optional component to perform a function x, such as debugging or testing), and a micro-controller **48**. During power saving mode a majority of the circuitry of the crossbar **44**, PLL and SerDes **46** are powered down saving around 80-95% of the power used by the retimer device. Enough of the circuitry of the retimer integrated circuit **44** is left powered up in order to allow the retimer device **12** to detect the wakeup signal described in more detail above and optionally send a wakeup signal to another device on the active data link **14**.

(30) Various features of the invention which are, for clarity, described in the contexts of separate embodiments may also be provided in combination in a single embodiment. Conversely, various features of the invention which are, for brevity, described in the context of a single embodiment may also be provided separately or in any suitable sub-combination.

(31) The embodiments described above are cited by way of example, and the present invention is not limited by what has been particularly shown and described hereinabove. Rather the scope of the invention includes both combinations and sub-combinations of the various features described hereinabove, as well as variations and modifications thereof which would occur to persons skilled in the art upon reading the foregoing description and which are not disclosed in the prior art.

Claims

1. A retimer device, comprising: a receiver to receive data from a first device via a data link; retimer circuitry to: recover a clock phase from the received data; and prepare for transmission a new copy of the received data sampled by a clean clock based on the recovered clock phase; and a transmitter to transmit the new copy of the received data to a second device via the data link, wherein the receiver is configured to: receive an in-band standby signal from the first device having a given pattern in a physical layer of the signal; activate a power saving mode of the retimer device so that at least some of the retimer circuitry is powered down responsively to the standby signal having the given pattern in the physical layer of the standby signal; receive an in-band wakeup signal from the first device; and initiate an exit from the power saving mode to power up the retimer device responsively to the wakeup signal.

2. The device according to claim 1, wherein the receiver is configured to: activate the power saving mode of the retimer circuitry so that the retimer circuitry is powered down responsively to the in-band standby signal; and initiate an exit from the power saving mode to power up the retimer

circuitry responsively to the wakeup signal.

3. The device according to claim 1, wherein the receiver recognizes the given pattern in the standby signal in a physical media attachment sub layer without needing to use a physical coding sub layer.

4. The device according to claim 1, wherein the transmitter is configured to send another given pattern in the physical layer of another in-band standby signal to the second device to activate a power saving mode in the second device.

5. The device according to claim 1, wherein the transmitter is configured to send the wakeup signal in-band to the second device responsively to receiving the wakeup signal from the first device.

6. The device according to claim 5, wherein the transmitter is configured to send the wakeup signal to the second device prior to, or while, the retimer device is powering up from the power saving mode.

7. The device according to claim 1, wherein the transmitter is configured to generate another wakeup signal and send the other wakeup signal in-band to the second device responsively to receiving the wakeup signal from the first device.

8. The device according to claim 1, wherein the data link includes an electrical signal data link.

9. The device according to claim 1, wherein the data link includes an optical signal data link.

10. The device according to claim 1, wherein the first device is an end device, and the second device is an end device.

11. The device according to claim 1, wherein the first device is a retimer device and the second device is an end device.

12. The device according to claim 1, wherein the first device is an end device, and the second device is a retimer device.

13. The device according to claim 1, wherein the first device is a retimer device and the second device is a retimer device.

14. A retimer device power saving mode method, comprising: receiving data from a first device via a data link; recovering a clock phase from the received data; preparing for transmission a new copy of the received data sampled by a clean clock based on the recovered clock phase; transmitting the new copy of the received data to a second device via the data link; receiving an in-band standby signal from the first device having a given pattern in a physical layer of the signal; activating a power saving mode of a retimer device so that at least some of the retimer circuitry is powered down responsively to the standby signal having the given pattern in the physical layer of the standby signal; receiving an in-band wakeup signal from the first device; and initiating an exit from the power saving mode to power up the retimer device responsively to the wakeup signal.

15. The method according to claim 14, wherein: the activating includes activating the power saving mode of retimer circuitry so that the retimer circuitry is powered down responsively to the in-band standby signal; and initiating an exit from the power saving mode to power up the retimer circuitry responsively to the wakeup signal.

16. The method according to claim 14, further comprising recognizing the given pattern in the standby signal in a physical media attachment sub layer without needing to use a physical coding sub layer.

17. The method according to claim 14, further comprising sending another given pattern in the physical layer of another in-band standby signal to the second device to activate a power saving mode in the second device.

18. The method according to claim 14, further comprising sending the wakeup signal in-band to the second device responsively to receiving the wakeup signal from the first device.

19. The method according to claim 18, wherein the sending includes sending the wakeup signal to the second device prior to, or while, the retimer device is powering up from the power saving mode.

20. The method according to claim 14, further comprising generating another wakeup signal and sending the other wakeup signal in-band to the second device responsively to receiving the wakeup

signal from the first device.

21. The method according to claim 14, wherein the data link includes an electrical signal data link.

22. The method according to claim 14, wherein the data link includes an optical signal data link.

23. The method according to claim 14, wherein the first device is an end device, and the second device is an end device.

24. The method according to claim 14, wherein the first device is a retimer device and the second device is an end device.

25. The method according to claim 14, wherein the first device is an end device, and the second device is a retimer device.

26. The method according to claim 14, wherein the first device is a retimer device and the second device is a retimer device.
